

Simon Puthur Binu

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SUMMARY

Computer Science undergraduate with hands-on experience in software and electronics, including full-stack web and mobile development and Arduino-based electrical projects. Proficient in integration with React, Flutter, Firebase, Google Cloud, and Gemini AI. Hackathon winner with a strong track record of exploring and building practical solutions in various domains.

EDUCATION

Model Engineering College <i>Bachelor of Technology in Computer Science and Engineering</i>	Ernakulam, Kerala 2024 – 2028
Devamatha CMI Public School <i>Class X and XII</i>	Thrissur, Kerala 2012 – 2024

PROJECTS

Ravana <i>Node.js, socket.io, three.js</i>	February 2026
- Developed a decentralized rendering engine that parallelizes 3D path-tracing tasks across a swarm of mobile browsers, drastically reducing render times for complex scenes - Architected a high-performance binary pipeline using Float32Arrays and serialized BVH data to stream raw geometry directly to workers, bypassing the overhead of traditional file parsing - Engineered a real-time orchestration layer with Node.js and Socket.io to manage dynamic task queues and "stitch" incoming image tiles into a high-resolution final output - Collaborated in a 3-member team to deliver a functional "zero-install" prototype within a 20-hour hackathon window, focusing on rapid orchestration and hardware-agnostic compute - Winner of CUSAT's Make-A-Ton 8.0 Hackathon	
Sakshya <i>React, Firebase, Gemini, Google Cloud</i>	January 2026 – Present
- Developing a full-stack web platform for video analysis using React on the frontend and Firebase as a Backend-as-a-Service - Integrated Gemini 2.5 Flash and Pro models via Vertex AI to analyze compressed video content efficiently - Designed workflows to preprocess, compress, and analyze videos while optimizing for latency and cost - Collaborated in a 3-member team following rapid prototyping and iterative development practices - Winner of the Google Developer Group on Campus MEC Techsprint Hackathon 2026	
Medisukham <i>Flutter, Gemini, Google Cloud</i>	November 2025 – Present
- Building a mobile application to digitize medical prescriptions and automate medication reminders - Leveraged Gemini models to extract, infer, and structure dosage timings from unstructured prescription data - Implemented automated alarm scheduling based on inferred medication instructions - Designed a simple, accessible user interface optimized for older users - Focused on reliability and ease of use in a real-world healthcare-oriented use case	
Partify <i>discord.js, socket.io, Node.js</i>	March 2022 – July 2022
- Designed and built a real-time group music listening platform for Discord users, enabling synchronized playback across clients - Developed a Discord bot using discord.js to manage user sessions, commands, and room coordination - Implemented low-latency, bi-directional communication between the Discord bot and web client using socket.io - Integrated YouTube audio extraction to stream and share music seamlessly among multiple users	

TECHNICAL SKILLS

Languages: Python, Java, Javascript, C, C#, C++, GDScript, Dart
Frameworks and Software: React, Node.js, Flask, Svelte, FastAPI, Flutter, Godot Engine
Developer Tools: Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ
Languages: English (Professional Working Proficiency), Malayalam (Native), German (Elementary)
Interests: Web and App Development, Open Source Contribution, UI / UX Designing, Blogging, Game Development, Arduino Hardware Tinkering